Maulana Azad National Institute of Technology, Bhopal

Subject: PHYSICS (PHY-102)

Subject Coord.: Dr. Vasudev Thakre
A.P. Department of Physics
MANIT, Bhopal

Solid State Physics

Free electron theory of Metals :-

The main characteristics properties of metals are their high electrical and thermal conductivities.

Lorentz and Drude considered that about 95% of whole volume of solid is occupied by the electrons and only 5% of the volume is occupied by ion cores. ie. solids act as free e^- gas.

The pressure exerted by ~~free~~ the e^- gas at temp. T. acc. to classical theory is

$$P = \frac{1}{3} \rho (\bar{c})^2$$

where

$\bar{c}$ = rms velocity of e^-

ρ - density

for molar volume of metal

for molar volume of metal (ρ - density

density, $$\rho = \frac{M_A}{V_m} \quad \text{—— (1)}$$

$$\therefore P = \frac{1}{3} \frac{M_A}{V_m} \cdot \bar{c}^2 \quad \text{—— (2)}$$

where M_A is mass of N_A atom and V_m - volume (molar) ($M_A = mN_A$)

$$PV_m = \frac{1}{3} m \cdot N_A \bar{c}^2$$

$$\because PV_m = RT$$

$$RT = \frac{1}{3} m N_A \bar{c}^2$$

$$m\bar{c}^2 = \frac{3RT}{N_A} = 3T \cdot \frac{R}{N_A}$$

$$m\bar{c}^2 = 3k_B T \quad \text{—— (2)}$$

R - univ. gas const.

$k_B = \frac{R}{N_A}$ Boltzmann const.

N_A - Avogadro's No.

$\because$ The K.E. of e^- is.

$$\frac{1}{2} m \bar{c}^2 = \frac{3}{2} k_B T.$$

$$\bar{c}^2 = \frac{3 k_B T}{m}$$

$$\bar{c} = \sqrt{\frac{3 k_B T}{m}} \quad \text{—— (3)}$$

Ohm's law, Electrical Conductivity and Resistivity :-

we know that ohm's law

$$V = IR$$

$$R = \frac{V}{I} \quad \text{—— (4)}$$

as $R = \rho \frac{l}{A}$ & $\rho = \frac{1}{\sigma}$ —(5)

from (4) & (5)

$$\frac{V}{I} = \rho \frac{l}{A}$$

$$\frac{V}{I} = \frac{1}{\sigma} \cdot \frac{l}{A}$$

$$\sigma = \frac{I}{V} \cdot \frac{l}{A} = \frac{I}{A} \cdot \frac{l}{V}$$

$\because J = \frac{I}{A}$

$E = \frac{V}{l}$

$$\boxed{\sigma = J/E}$$

$$\boxed{J = \sigma E} \quad \text{—— (6)}$$

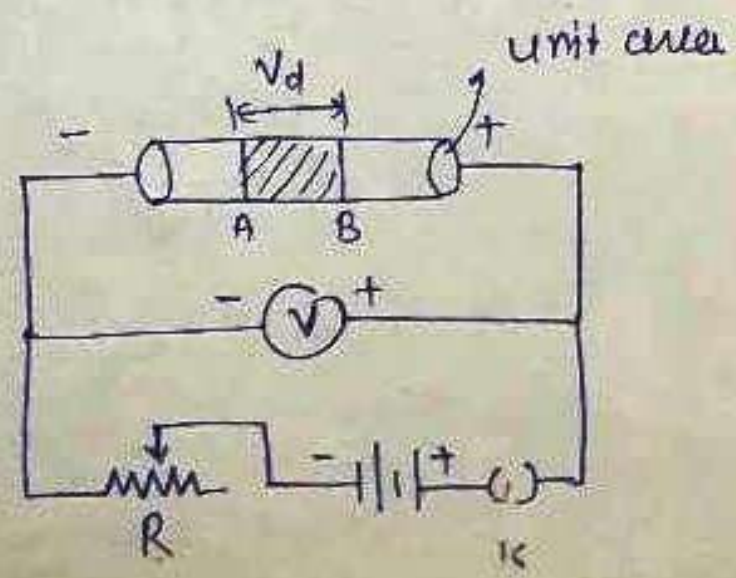
This is ohm's law in terms of conductivity.

* Conductivity (σ) :-

$\because$ $F = eE$ & $F = ma$

$ma = eE$

acceleration $a = \frac{eE}{m}$ —— (7)



Now considering the collision of e^- with an ion core.
let. λ = mean free path, then mean time between collision is $\tau = \frac{\lambda}{\bar{c}}$

when electric field is applied. then drift velocity

$$a = \frac{V_d}{\tau} \quad V_d = a\tau = \frac{eE}{m}\tau \quad \text{——} \quad (8)$$

$\because$ current density $J = neV_d = \frac{ne^2E\tau}{m}$

$\because$ ohm's law $J = \sigma E$

$$\sigma = \frac{J}{E} = \frac{ne^2\tau}{m}$$

$$\boxed{\sigma = \frac{ne^2\tau}{m}} \quad \text{——} \quad (9)$$

This is expression for electrical conductivity.

* <u>Resistivity</u>:

$$\because \quad \tau = \frac{\lambda}{\bar{c}}$$

$$\sigma = \frac{ne^2}{m} \cdot \frac{\lambda}{\bar{c}}$$

from eqn (3)

$$\sigma = \frac{ne^2\lambda}{m} \cdot \left[\frac{m}{3k_BT}\right]^{\frac{1}{2}}$$

$$\boxed{\sigma = \frac{ne^2\lambda}{\sqrt{3mk_BT}}} \quad \text{--- (10)}$$

or resistivity $\rho = \frac{1}{\sigma}$

$$\boxed{\rho = \frac{\sqrt{3mk_BT}}{ne^2\lambda}} \quad \text{--- (11)}$$

i.e $\boxed{\rho \propto \sqrt{T}}$

Fermi - Dirac Statistical Distribution Function:-

Electrons in solids obey Fermi- Dirac statistics. This statistics results that the distribution of electrons over a range of allowed energy levels at thermal equilibrium is

$$\boxed{F(E) = \frac{1}{1 + e^{(E - E_F)/KT}}} \quad \text{——— (1)}$$

where K = Boltzmann's constant $(= 1.38 \times 10^{-23}\ J/K)$
$(= 8.62 \times 10^{-5}\ eV/K)$

E_F = Fermi energy and.

T = absolute temp.

" Fermi- Dirac distribution function gives the probability that an available energy state at E will be occupied by an e^- at absolute temp. T."

Case-I If $E = E_F$ and $T \neq 0K$, $\frac{E-E_F}{KT} = 0$

$$f(E) = \frac{1}{e^0+1} = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

i.e. probability = 50%

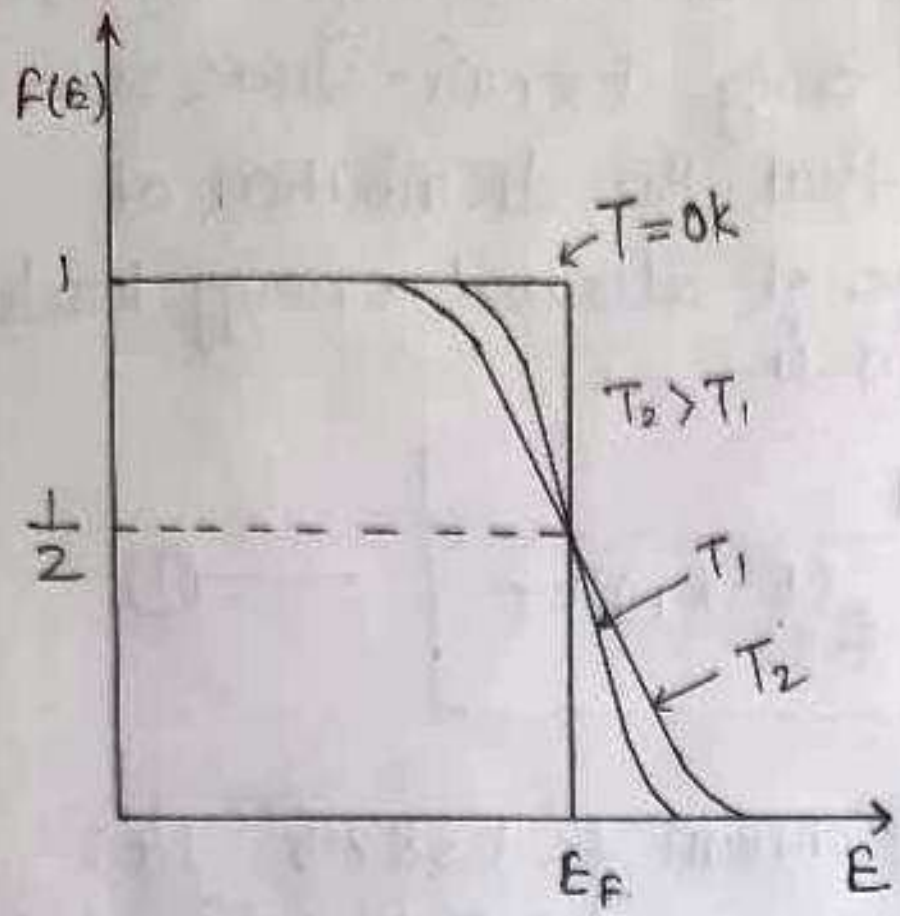
Case-II If $E > E_F$ and $T = 0K$, $\frac{E-E_F}{KT} = +\infty$

$$\Rightarrow \quad f(E) = \frac{1}{e^{+\infty}+1} = \frac{1}{\infty+1} = \frac{1}{\infty} = 0$$

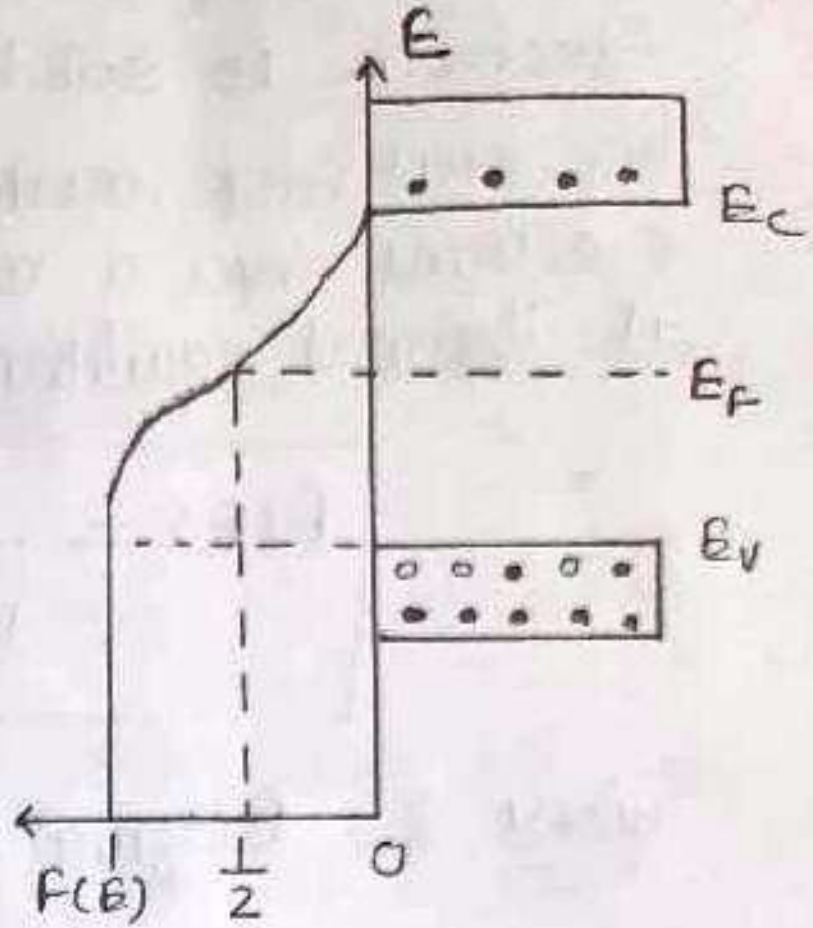
ie Probability = 0%.

Case-III If $E < E_F$ and $T = 0K$, $\frac{E-E_F}{KT} = -\infty$

$$\Rightarrow \quad f(E) = \frac{1}{e^{-\infty}+1} = \frac{1}{0+1} = 1$$ ie Prob. = 100%



(a) The Fermi-Dirac distribution function



(b) Intrinsic Semi-conductor.

Intrinsic Semiconductor :- The semiconductors in which the transformation of e^- to the conduction band and the generation of holes in the valence band are achieved purely by thermal excitation are called intrinsic semiconductors. (Eg. Ge & Si)

Electron Concentration in Conduction Band :-

For this we assume the following:

(i) Electrons in conduction band behave as free particles with an effective mass m_e^*.

(ii) The number of conduction e^- per cubic meter whose energies lie betn. E and $E+dE$ is given by

$$dn = N(E) \cdot f(E) dE \quad --- (1)$$

Where- $N(E)$ - density of states at the bottom of C.B. and

$$N(E) = \frac{4\pi}{h^3} \cdot (2m_e^*)^{3/2} \cdot (E-E_c)^{1/2}$$

Here E_c - Energy at the bottom of C.B.

and $$f(E) = \frac{1}{1+e^{(E-E_F)/kT}}$$

Since the e^- in the C.B. may have energies lying betn. E_c and ∞. the concentration of e^- in the C.B. is given by

$$n = \int^{\infty} N(E) \cdot f(E)\, dE$$

$$n = \int_{E_c}^{\infty} \frac{4\pi}{h^3} \cdot (2m_e^*)^{3/2} \frac{(E-E_c)^{1/2}}{1+e^{(E-E_F)/KT}} dE$$

$$n = \frac{4\pi}{h^3} \cdot (2m_e^*)^{3/2} \int_{E_c}^{\infty} \frac{(E-E_c)^{1/2}}{1+e^{(E-E_F)/KT}} dE$$

for $E \geq E_c$ and $E - E_F >> KT$.

$$1 + e^{(E-E_F)/KT} \approx e^{(E-E_F)/KT}$$

$$n = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_c}^{\infty} \frac{(E-E_c)^{1/2}}{e^{(E-E_F)/KT}} dE$$

$$n = \frac{4\pi}{h^3} \cdot (2m_e^*)^{3/2} \int_{E_c}^{\infty} (E-E_c)^{1/2} e^{-(E-E_F)/KT} \cdot dE$$

$$n = \frac{4\pi}{h^3} \cdot (2m_e^*)^{3/2} e^{(E_F - E_c)/kT} \int_{E_c}^{\infty} (E - E_c)^{1/2} e^{-(E-E_c)/kT} dE$$

let $(E - E_c)/kT = x \Rightarrow dE = kT\,dx$ then

$$n = \frac{4\pi}{h^3} \cdot (2m_e^*)^{3/2} e^{(E_F - E_c)/kT} \int_0^{\infty} x^{1/2} (kT)^{1/2} e^{-x} kT\,dx$$

$$= \frac{4\pi}{h^3} \cdot (2m_e^*)^{3/2} (kT)^{3/2} e^{(E_F - E_c)/kT} \int_0^{\infty} x^{1/2} \cdot e^{-x} dx$$

$$= \frac{4\pi}{h^3} \cdot (2m_e^* kT)^{3/2} \cdot e^{(E_F - E_c)/kT} \cdot \int_0^{\infty} x^{1/2} \cdot e^{-x} dx$$

$$\because \int_0^\infty x^{1/2} e^{-x} dx = \frac{\sqrt{\pi}}{2}$$

$$n = \frac{4\pi}{h^3} (2m_e^* kT)^{3/2} e^{(E_F - E_c)/kT} \frac{\sqrt{\pi}}{2}$$

$$n = 2 \left[\frac{2\pi m_e^* kT}{h^2} \right]^{3/2} e^{(E_F - E_c)/kT}$$

$$\boxed{n = N_c \cdot e^{(E_F - E_c)/kT}} \quad \text{——— (1)}$$

where $N_c = 2 \left\{ \frac{2\pi m_e^* kT}{h^2} \right\}^{3/2}$

Hole Concentration In Valence Band:-

If $F(E)$ is probability for occupancy of an energy state E by an e^- then the probability that the energy state is vacant is given by $\{1-F(E)\}$ (ie. fermi function for holes).

$$\because \quad 1-f(E) = 1 - \frac{1}{1+e^{(E-E_F)/KT}}$$

$$= 1 - \{1+e^{(E-E_F)/KT}\}^{-1}$$

$$= 1 - \{1-e^{(E-E_F)/KT}\}$$

for $\frac{E-E_F}{KT} \ll 1$

$$1-f(E) = e^{(E-E_F)/KT} \quad \text{——} \quad (1)$$

for the top of the V.B. the density of states is given by

$$N(E) = \frac{4\pi}{h^3}(2m_h^*)^{3/2}(E_v - E)^{1/2}$$

where $\Rightarrow$ m_h^* - is effective mass of hole

E_v - Energy at top of V.B.

$\therefore$ Density of holes in the valence band (V.B) is given by.

$$P = \int_{-\infty}^{E_v} N(E)\{1 - f(E)\}dE$$

$$P = \frac{4\pi}{h^3}\cdot(2m_h^*)^{3/2}\int_{-\infty}^{E_v}(E_v - E)^{1/2}\cdot e^{(E-E_F)/kT}\cdot dE$$

$$P = \frac{4\pi}{h^3} \cdot (2m_h^*)^{3/2} \cdot e^{(E_V - E_F)/KT} \int_{-\infty}^{E_V} (E_V - E)^{1/2} e^{(E - E_V)/KT} \cdot dE$$

$$P = \frac{4\pi}{h^3} \cdot (2m_h^*)^{3/2} e^{(E_V - E_F)/KT} \cdot \int_{-\infty}^{E_V} (E_V - E)^{1/2} e^{-(E_V - E)/KT} \cdot dE$$

let $\frac{E_V - E}{KT} = x \Rightarrow dE = -KT\,dx$

$$P = \frac{4\pi}{h^3} \cdot (2m_h^*)^{3/2} \cdot e^{(E_V - E_F)/KT} \cdot \int_{+\infty}^{0} x^{1/2} (KT)^{1/2} e^{-x} KT(-dx)$$

or

$$P = \frac{4\pi}{h^3} \cdot (2m_h^*)^{3/2} \cdot (KT)^{3/2} e^{(E_V - E_F)/KT} \cdot \int_{0}^{+\infty} x^{1/2} e^{-x} dx$$

$$\because \int_0^{\infty} x^{1/2} e^{-x} dx = \frac{\sqrt{\pi}}{2}$$

$$\because \int_0^\infty x^{1/2} e^{-x} dx = \frac{\sqrt{\pi}}{2}$$

$$P = \frac{4\pi}{h^3} \cdot (2m_h^*)^{3/2} (KT)^{3/2} \cdot e^{(E_V - E_F)/KT} \cdot \frac{\sqrt{\pi}}{2}$$

$$P = 2 \left\{ \frac{2\pi m_h^* KT}{h^2} \right\}^{3/2} e^{(E_V - E_F)/KT}$$

or $$\boxed{P = N_V \cdot e^{(E_V - E_F)/KT}}$$

where $$N_V = 2 \left\{ \frac{2\pi m_h^* KT}{h^2} \right\}^{3/2}$$

Fermi Level :- The highest occupied energy level is called the fermi level and the energy corresponding to the level is called Fermi Energy (E_F).

Fermi level in intrinsic semiconductor :-

In intrinsic semiconductor, the free e^- and hole concentrations are equal ie

$$n = p.$$

$$\therefore \quad N_c e^{-(E_c - E_F)/KT} = N_v e^{-(E_F - E_v)/KT} \quad \text{--- (1)}$$

where N_c & N_v = effective density of states in C.B & V.B.

$$N_c = 2\left\{\frac{2\pi m_n^* KT}{h^2}\right\}^{3/2} \qquad N_v = 2\left\{\frac{2\pi m_p^* KT}{h^2}\right\}^{3/2}$$

taking log (eqn (1)) on both sides. from (1)

$$e^{-(E_c - E_F)/KT} = \frac{N_v}{N_c} e^{-(E_F - E_v)/KT}$$

taking log both sides-

$$-\frac{(E_c - E_f)}{KT} = \log\frac{N_v}{N_c} - \frac{(E_f - E_v)}{KT}$$

$$-E_c + E_f = KT\log\frac{N_v}{N_c} - E_f + E_v$$

$$E_f + E_f = KT\log\frac{N_v}{N_c} + E_c + E_v$$

$$2E_f = KT\log\frac{N_v}{N_c} + (E_c + E_v)$$

$$E_f = \frac{KT}{2}\log\frac{N_v}{N_c} + \frac{(E_c + E_v)}{2} \quad \text{—— (1)}$$

Substituting N_c & N_v, we get.

$$E_f = \frac{KT}{2}.\log\left(\frac{m_p^*}{m_n^*}\right)^{3/2} + \frac{(E_c + E_v)}{2} \quad \text{—— (2)}$$

If $m_p^* = m_n^*$ then

$$E_f = \frac{E_c + E_v}{2}$$

$$E_f = \frac{E_c + E_v + E_v - E_v}{2} = \frac{E_c - E_v}{2} + \frac{\cancel{2}E_v}{\cancel{2}}$$

$$E_f = \frac{E_c - E_v}{2} + E_v \qquad \because E_c - E_v = E_g$$

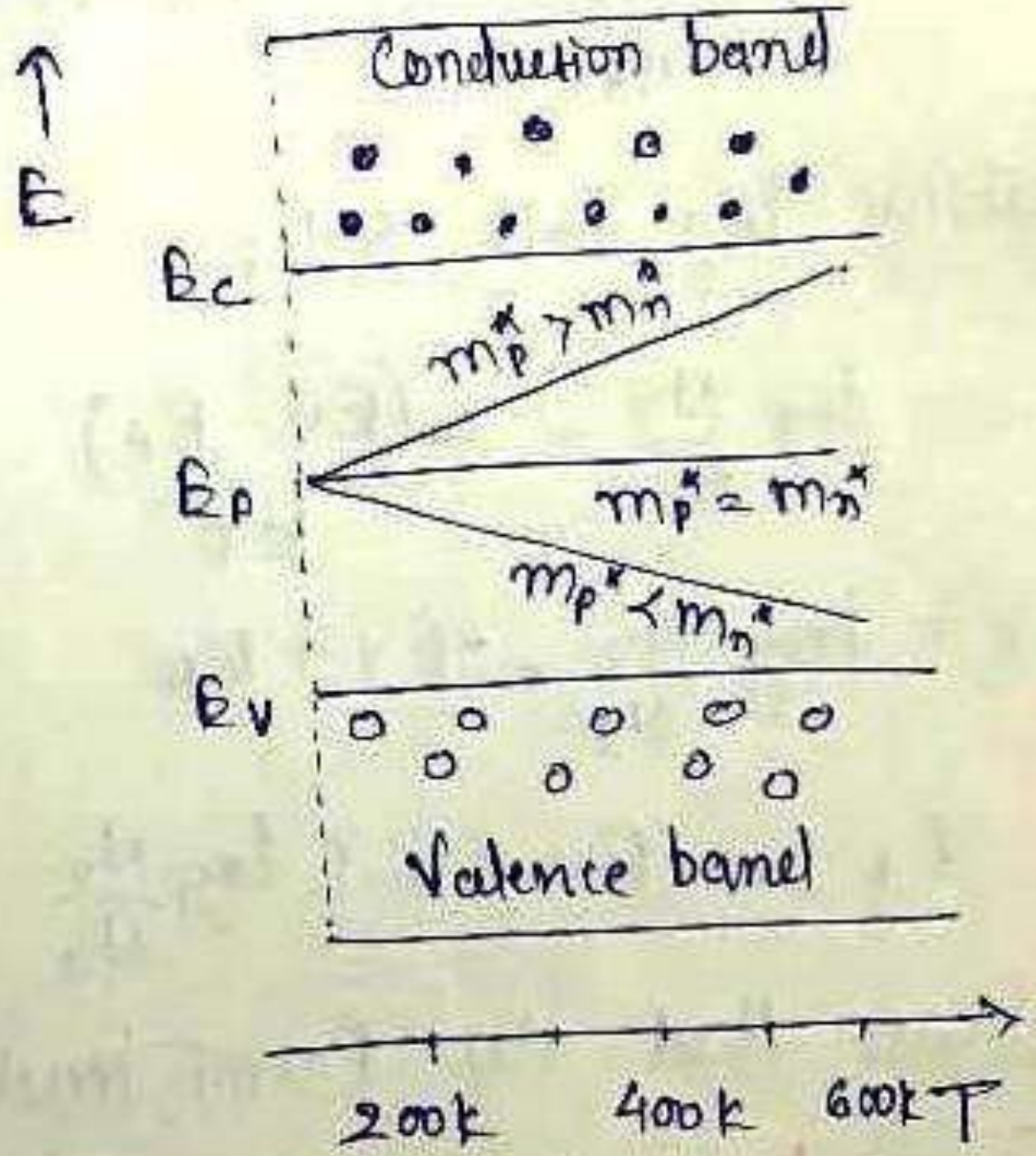
$$E_f = \frac{E_g}{2} + E_v$$

If the top of the valence band E_v is taken as zero level

ie $\boxed{E_v = 0}$ then

$$\boxed{E_f = \frac{E_g}{2}} \quad \text{——— (3)}$$

ie. Fermi energy is that energy for which Fermi-Dirac function equals 0.5 and it is the average energy possessed by charge carriers in a solid.


Conduction band
E
Ec
$m_p^* > m_n^*$
Ep
$m_p^* = m_n^*$
$m_p^* < m_n^*$
Ev
Valence band
200k
400k
600k
T

Fermi level in Extrinsic semiconductor:-

The conductivity of an intrinsic semiconductor can be increased significantly by adding certain impurities to it. The impure semiconductor is called extrinsic semiconductor. Extrinsic semiconductors are two type: (I) P-type S.C.

(II) N-type S.C.

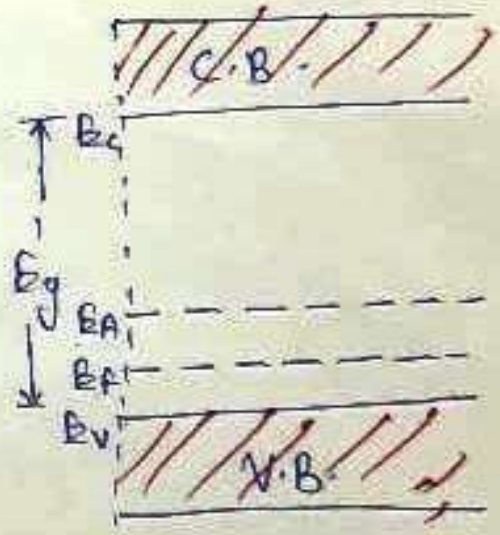
1.) Fermi level in P-type semiconductor:-

If we assume there are only acceptor atoms present and that these are all ionised, then $p = N_a$ (density of acceptor atoms) then

$$P = N_a = 2\left\{\frac{2\pi m_h^* kT}{h^2}\right\}^{3/2} \cdot e^{-(E_F - E_V)/kT}$$

$$N_a = N_V \, e^{-(E_V - E_F)/kT}$$

$$\therefore \quad \frac{N_v}{N_a} = e^{-(E_v - E_F)/KT}$$



taking log both sides-

$$\log \frac{N_v}{N_a} = -\frac{(E_v - E_F)}{KT}$$

$$KT \log \frac{N_v}{N_a} = -E_v + E_F$$

or $$E_F = E_v + KT \log \frac{N_v}{N_a}$$

It shows that the fermi level lies above the top of valence band.

2) Fermi level in N-type semiconductor:-

If we assume there are only donor atoms are present and then density of e^- is equal to density of donor atoms-

i.e $$n = N_d = 2\left\{\frac{2\pi m_n^* KT}{h^2}\right\}^{3/2} \cdot e^{-(E_c - E_F)/KT}$$

$$N_d = N_c \cdot e^{(E_F - E_c)/KT}$$

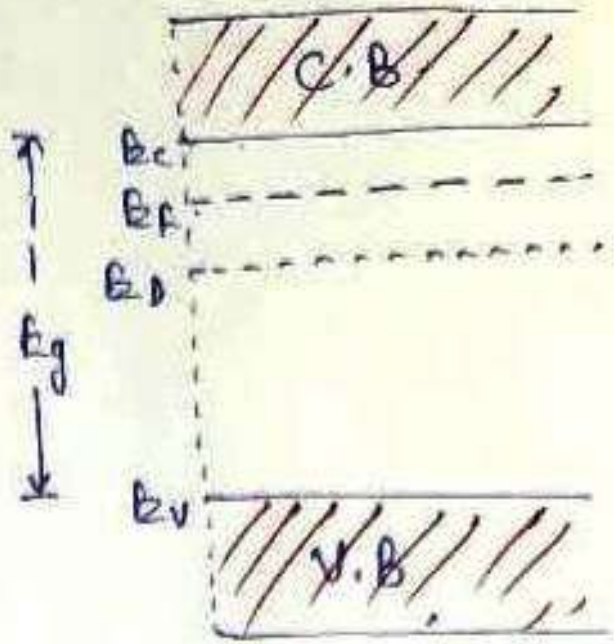
$$\therefore \quad \frac{N_c}{N_d} = e^{-(E_F - E_c)/KT}$$

taking log both sides we get

$$\log \frac{N_c}{N_d} = -\frac{(E_F - E_c)}{KT}$$

$$KT \log \frac{N_c}{N_d} = -E_F + E_c$$

$$E_F = E_c - KT \log \frac{N_c}{N_d}$$



It shows that the fermi level lies below the bottom of the conduction band.